Principios de la Robótica Autónoma

Filtro de Bayes – Filtro de Kalman

Ignacio Mas

Repaso de filtro de Bayes

$$Bel(x_t) = \eta\, p(z_t \mid x_t) \int p(x_t \mid u_t, x_{t-1}) Bel(x_{t-1}) dx_{t-1}$$

- Predicción

$$\overline{Bel}(x_t) = \int p(x_t \mid u_t, x_{t-1}) Bel(x_{t-1}) dx_{t-1}$$

- Corrección

$$Bel(x_t) = \eta\, p(z_t \mid x_t) \overline{Bel}(x_t)$$

Repaso de filtro de Bayes

$$Bel(x_t) = \eta \; P(z_t \mid x_t) \int P(x_t \mid u_t, x_{t-1}) \; Bel(x_{t-1}) \; dx_{t-1}$$

Algoritmo **Bayes_filter**(*Bel(x),d*):
η=0
If *d* is a perceptual data item *z* then
    For all *x* do
        $Bel'(x) = P(z \mid x) Bel(x)$
        $\eta = \eta + Bel'(x)$
    For all *x* do
        $Bel'(x) = \eta^{-1} Bel'(x)$
Else if *d* is an action data item *u* then
    For all *x* do
        $Bel'(x) = \int P(x \mid u, x') \; Bel(x') \; dx'$
Return *Bel'(x)*

Filtro de Kalman

- Filtro de Bayes con **Gaussianas**
- Desarrollado a fines de los 1950's
- Filtro de Bayes más utilizado en la práctica
- Usado en economía, pronóstico climático, navegación satelital, robótica, etc.

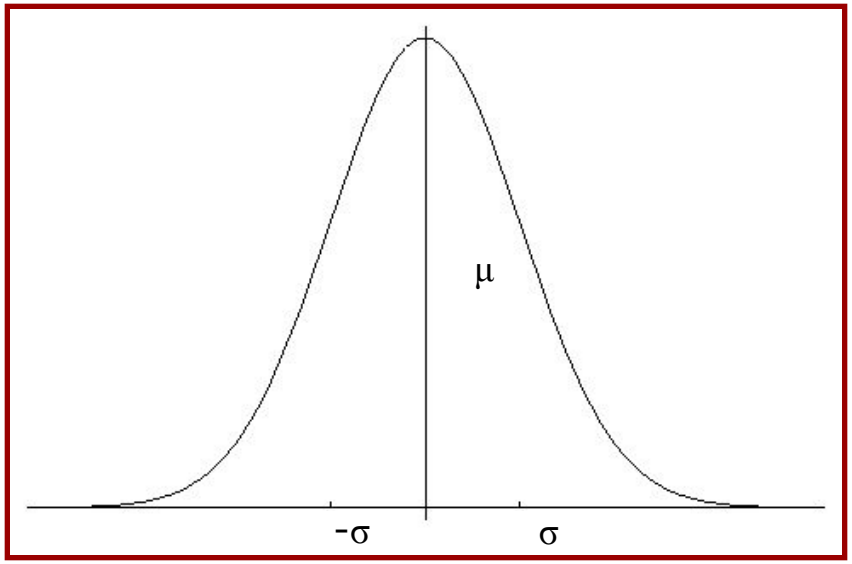
- El algoritmo del filtro de Kalman es sólo **multiplicar matrices**!

Gaussianas

$$p(x) \sim N(\mu, \sigma^2):$$

$$p(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\frac{(x-\mu)^2}{\sigma^2}}$$

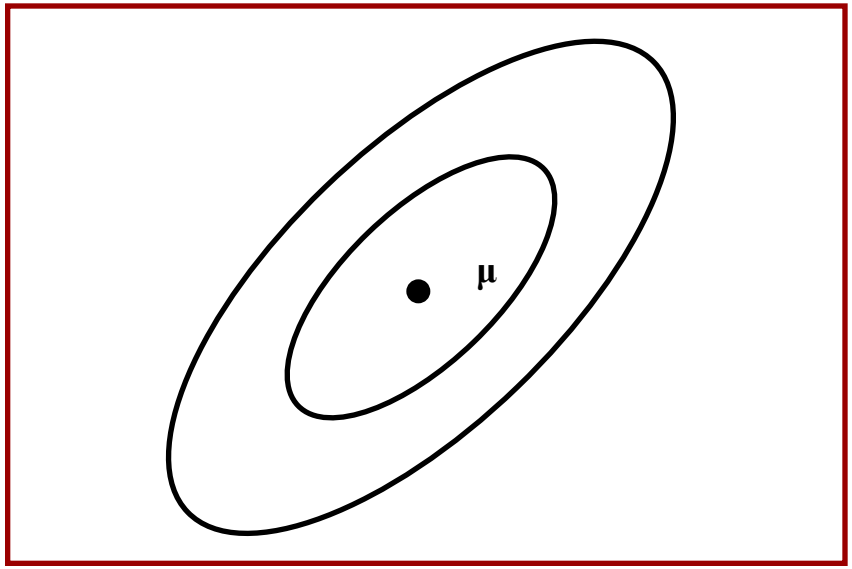
Una variable



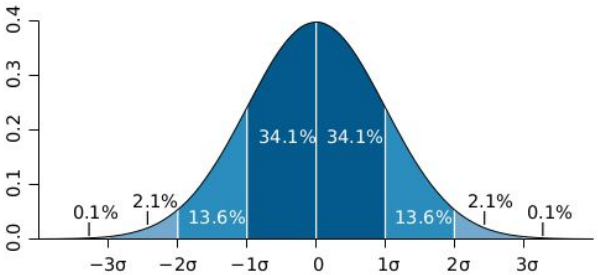
$$p(\mathbf{x}) \sim N(\mu, \Sigma):$$

$$p(\mathbf{x}) = \frac{1}{(2\pi)^{d/2} |\Sigma|^{1/2}} e^{-\frac{1}{2}(\mathbf{x}-\mu)^t \Sigma^{-1} (\mathbf{x}-\mu)}$$

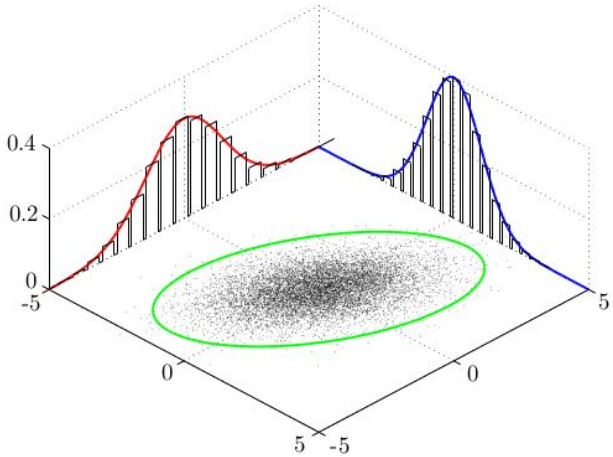
Multivariable



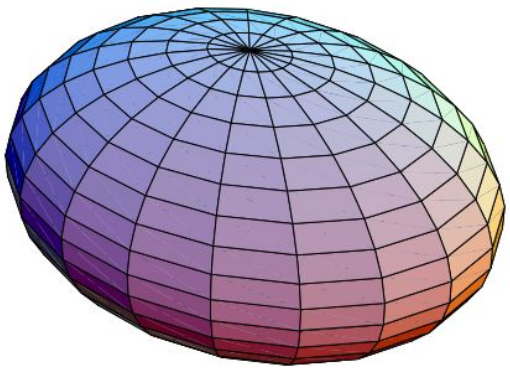
Gaussianas



1D



2D



3D

Propiedades de Gaussianas

- Caso univariable

$$\left.\begin{array}{l} X \sim N(\mu, \sigma^2) \\ Y = aX + b \end{array}\right\} \Rightarrow \quad Y \sim N(a\mu + b, a^2\sigma^2)$$

$$\left.\begin{array}{l} X_1 \sim N(\mu_1, \sigma_1^{\ 2}) \\ X_2 \sim N(\mu_2, \sigma_2^{\ 2}) \end{array}\right\} \Rightarrow p(X_1) \cdot p(X_2) \sim N\left(\frac{\sigma_2^{\ 2}}{\sigma_1^{\ 2} + \sigma_2^{\ 2}} \mu_1 + \frac{\sigma_1^{\ 2}}{\sigma_1^{\ 2} + \sigma_2^{\ 2}} \mu_2, \quad \frac{1}{\sigma_1^{\ -2} + \sigma_2^{\ -2}} \right)$$

Propiedades de Gaussianas

- Caso multivariable

$$\left.\begin{array}{l} X \sim N(\mu, \Sigma) \\ Y = AX + B \end{array}\right\} \Rightarrow \quad Y \sim N(A\mu + B, A\Sigma A^T)$$

$$\left.\begin{array}{l} X_1 \sim N(\mu_1, \Sigma_1) \\ X_2 \sim N(\mu_2, \Sigma_2) \end{array}\right\} \Rightarrow p(X_1) \cdot p(X_2) \sim N\left(\frac{\Sigma_2}{\Sigma_1 + \Sigma_2} \mu_1 + \frac{\Sigma_1}{\Sigma_1 + \Sigma_2} \mu_2, \quad \frac{1}{\Sigma_1^{-1} + \Sigma_2^{-1}} \right)$$

(donde la división "–" denota inversión de la matriz)

- Seguirá siendo **Gaussiana** si empezamos con Gaussianas y hacemos sólo **transformaciones afines y productos**

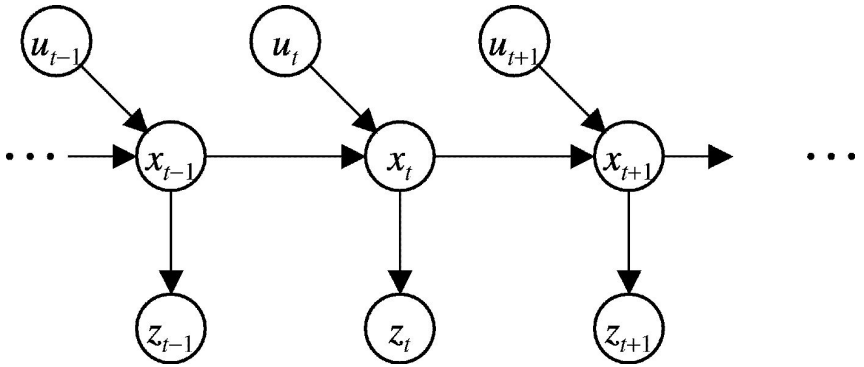
Filtro de Kalman Discreto

Estima el estado x de un proceso de tiempo discreto que es gobernado por una ecuación lineal estocástica en diferencias:

$$\boxed{x_t = A_t x_{t-1} + B_t u_t + \varepsilon_t}$$

con una medición:

$$\boxed{z_t = C_t x_t + \delta_t}$$



Componentes del filtro de Kalman

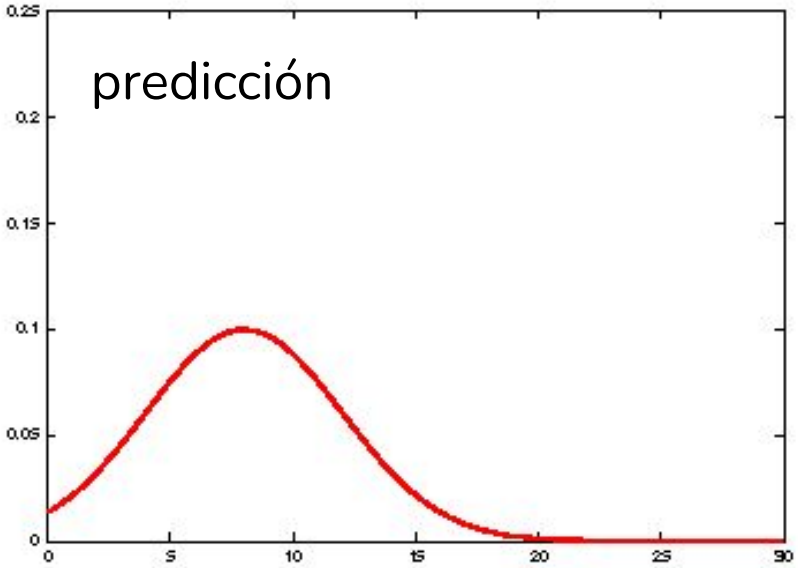
A_t Matriz ($n \times n$) que describe cómo el estado evoluciona de t-1 a t sin control ni ruido.

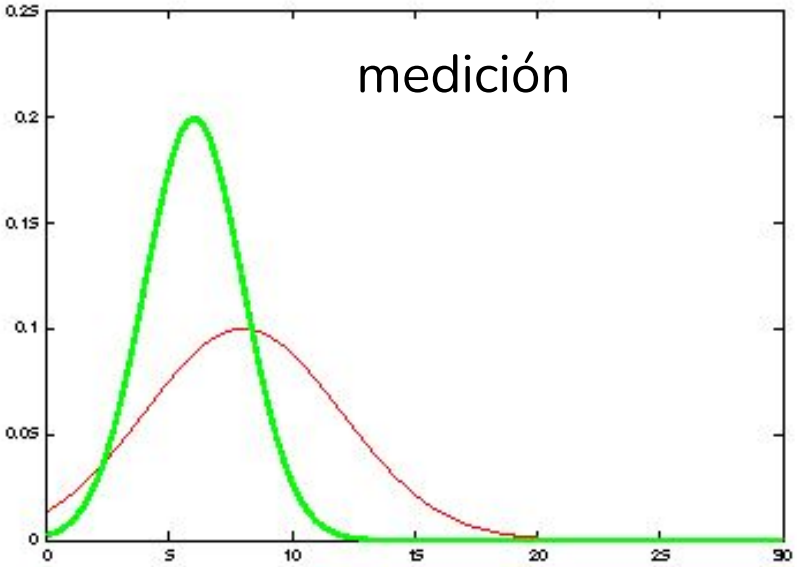
B_t Matriz ($n \times l$) que describe como el control u_t cambia el estado de t-1 a t.

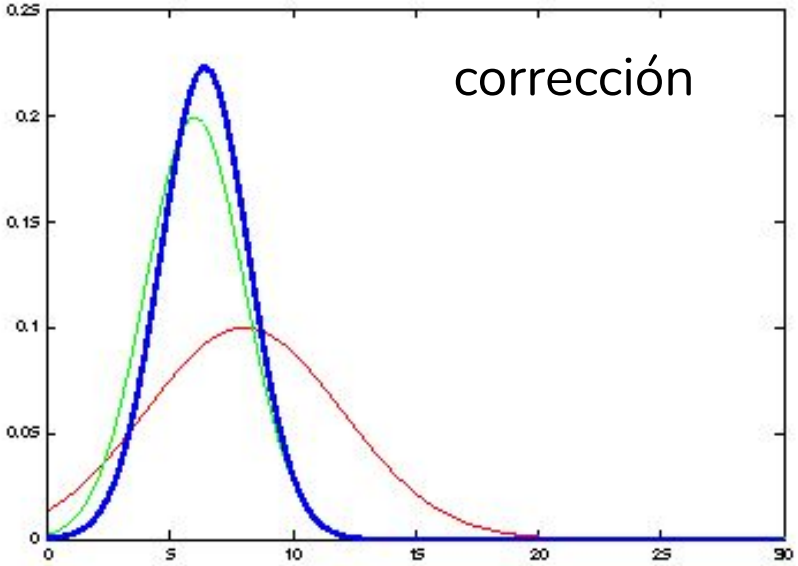
C_t Matriz ($k \times n$) que describe cómo mapear el estado x_t a una observación z_t.

ε_t δ_t Variables aleatorias que representan el ruido de proceso y de medición que se asumen independientes y normalmente distribuidas con covarianzas Q_t y R_t respectivamente.

Filtro de Kalman en 1D

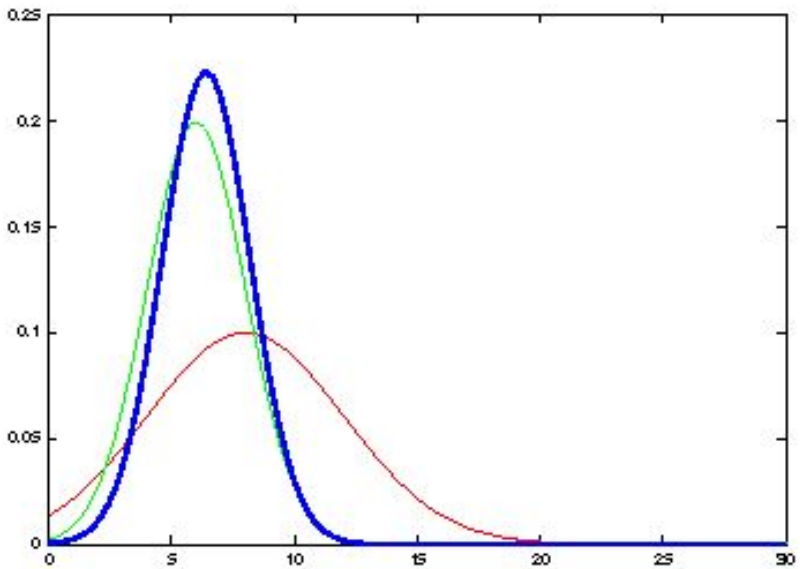






Es un promedio ponderado

Filtro de Kalman en 1D



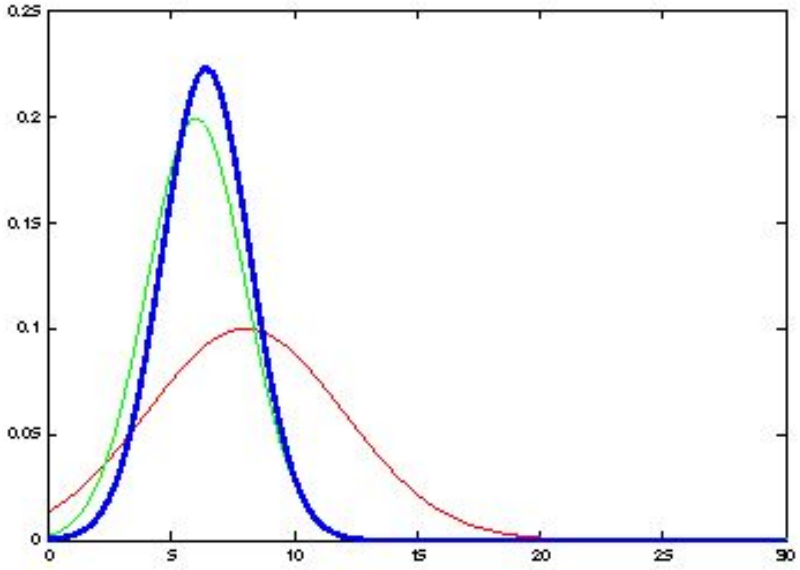
Cómo obtener la curva azul?

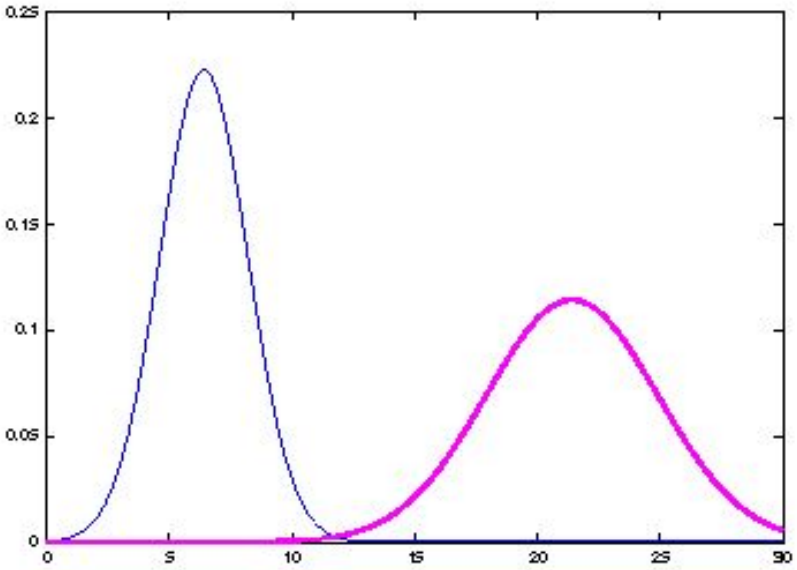
Paso de corrección del filtro de Kalman

$$bel(x_t) = \begin{cases} \mu_t = \bar{\mu}_t + K_t(z_t - C_t\bar{\mu}_t) \\ \sigma_t^2 = (1 - K_t C_t)\bar{\sigma}_t^2 \end{cases} \quad \text{con} \quad K_t = \frac{\bar{\sigma}_t^2}{\bar{\sigma}_t^2 + \bar{\sigma}_{obs,t}^2}$$

$$bel(x_t) = \begin{cases} \mu_t = \bar{\mu}_t + K_t(z_t - C_t\bar{\mu}_t) \\ \Sigma_t = (I - K_t C_t)\bar{\Sigma}_t \end{cases} \quad \text{con} \quad K_t = \bar{\Sigma}_t C_t^T (C_t \bar{\Sigma}_t C_t^T + R_t)^{-1}$$

Filtro de Kalman en 1D





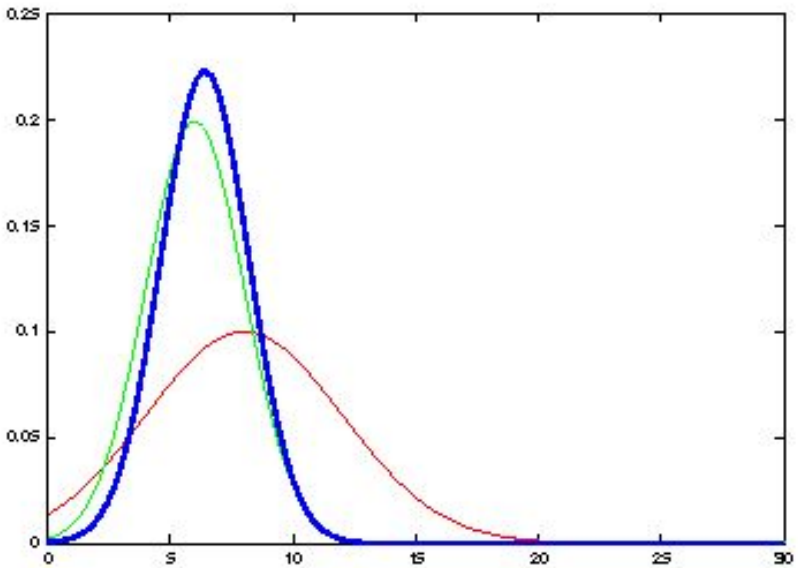
$$\overline{bel}(x_t) = \begin{cases} \bar{\mu}_t = a_t \mu_{t-1} + b_t u_t \\ \bar{\sigma}_t^2 = a_t^2 \sigma_t^2 + \sigma_{act,t}^2 \end{cases}$$

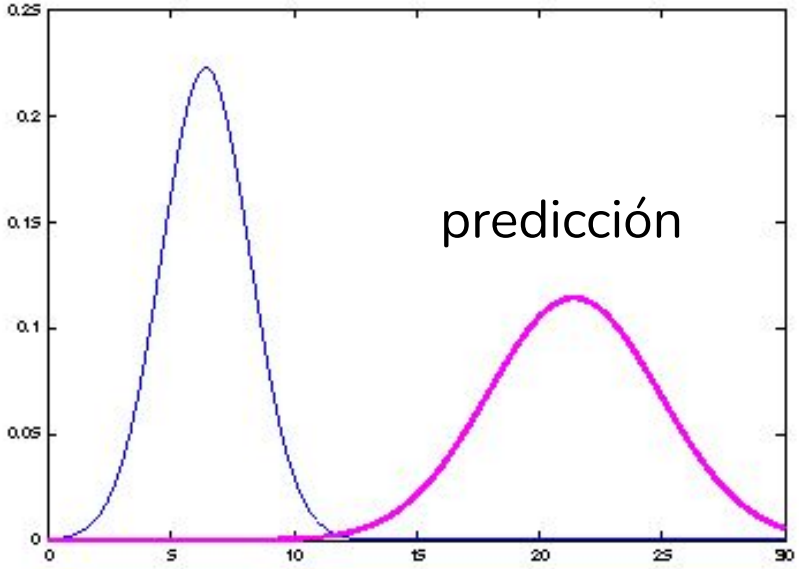
$$\overline{bel}(x_t) = \begin{cases} \bar{\mu}_t = A_t \mu_{t-1} + B_t u_t \\ \bar{\Sigma}_t = A_t \Sigma_{t-1} A_t^T + Q_t \end{cases}$$

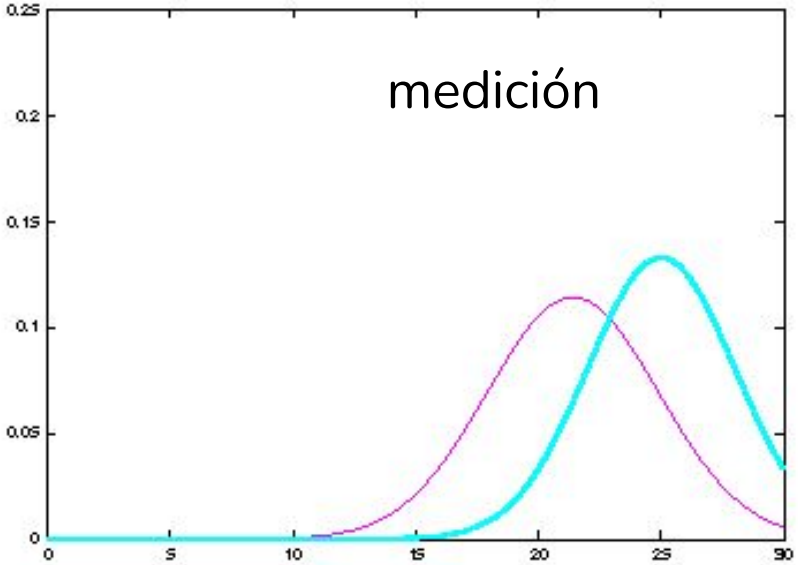
Cómo obtener la curva magenta?

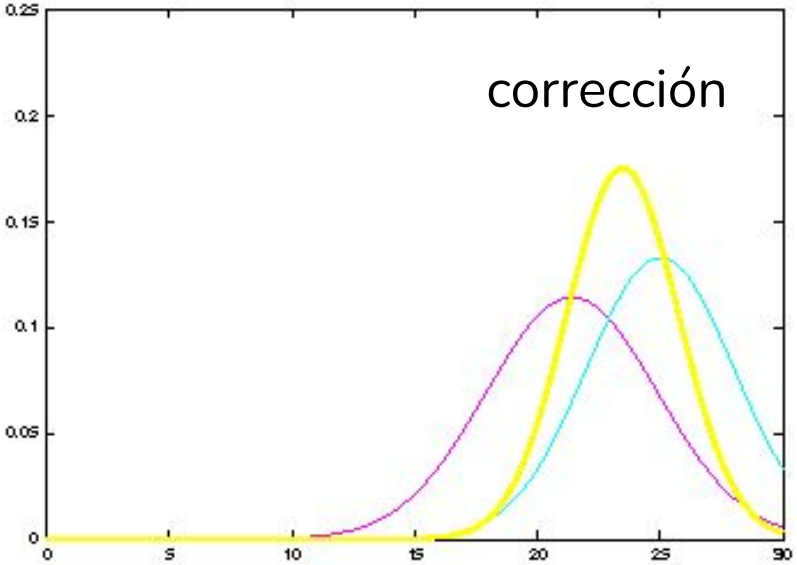
Paso de predicción de estado

Filtro de Kalman en 1D









Sistemas Gaussianos Lineales: Inicialización

El Belief inicial está normalmente distribuido:

$$\boxed{bel(x_0) = N\left(x_0; \mu_0, \Sigma_0\right)}$$

Sistemas Gaussianos Lineales: Dinámica

La dinámica son funciones **lineales** del estado y el control más ruido aditivo:

$$\boxed{x_t = A_t x_{t-1} + B_t u_t + \varepsilon_t}$$

$$\boxed{p(x_t \mid u_t, x_{t-1}) = N\left(x_t; A_t x_{t-1} + B_t u_t, Q_t\right)}$$

$$\boxed{\begin{array}{c} \overline{bel}(x_t) = \int p(x_t \mid u_t, x_{t-1})\, bel(x_{t-1}) dx_{t-1} \\ \Downarrow \qquad\qquad \Downarrow \\ \sim N\left(x_t; A_t x_{t-1} + B_t u_t, Q_t\right) \quad \sim N\left(x_{t-1}; \mu_{t-1}, \Sigma_{t-1}\right) \end{array}}$$

Sistemas Gaussianos Lineales: Dinámica

$$\overline{bel}(x_t) = \int p(x_t \mid u_t, x_{t-1}) \qquad bel(x_{t-1}) dx_{t-1}$$

$$\Downarrow \qquad\qquad \Downarrow$$

$$\sim N\left(x_t; A_t x_{t-1} + B_t u_t, Q_t\right) \sim N\left(x_{t-1}; \mu_{t-1}, \Sigma_{t-1}\right)$$

$$\Downarrow$$

$$\overline{bel}(x_t) = \eta \int \exp\left\{-\frac{1}{2}(x_t - A_t x_{t-1} - B_t u_t)^T Q_t^{-1} (x_t - A_t x_{t-1} - B_t u_t)\right\}$$

$$\exp\left\{-\frac{1}{2}(x_{t-1} - \mu_{t-1})^T \Sigma_{t-1}^{-1} (x_{t-1} - \mu_{t-1})\right\} dx_{t-1}$$

$$\overline{bel}(x_t) = \begin{cases} \overline{\mu}_t = A_t \mu_{t-1} + B_t u_t \\ \overline{\Sigma}_t = A_t \Sigma_{t-1} A_t^T + Q_t \end{cases}$$

Sistemas Gaussianos Lineales: Observaciones

Las observaciones son funciones lineales del estado, más ruido aditivo:

$$\boxed{z_t = C_t x_t + \delta_t}$$

$$\boxed{p(z_t \mid x_t) = N(z_t; C_t x_t, R_t)}$$

$$\boxed{\begin{array}{lcc} bel(x_t) = \eta & p(z_t \mid x_t) & \overline{bel}(x_t) \\ & \Downarrow & \Downarrow \\ & \sim N(z_t; C_t x_t, R_t) & \sim N(x_t; \overline{\mu}_t, \overline{\Sigma}_t) \end{array}}$$

Sistemas Gaussianos Lineales: Observaciones

$$bel(x_t) = \eta \quad p(z_t \mid x_t) \qquad \overline{bel}(x_t)$$

$$\Downarrow \qquad\qquad \Downarrow$$

$$\sim N(z_t; C_t x_t, R_t) \qquad \sim N(x_t; \overline{\mu}_t, \overline{\Sigma}_t)$$

$$\Downarrow$$

$$bel(x_t) = \eta \exp\left\{-\frac{1}{2}(z_t - C_t x_t)^T R_t^{-1}(z_t - C_t x_t)\right\} \exp\left\{-\frac{1}{2}(x_t - \overline{\mu}_t)^T \overline{\Sigma}_t^{-1}(x_t - \overline{\mu}_t)\right\}$$

$$bel(x_t) = \begin{cases} \mu_t = \overline{\mu}_t + K_t(z_t - C_t \overline{\mu}_t) \\ \Sigma_t = (I - K_t C_t)\overline{\Sigma}_t \end{cases} \qquad \text{with} \quad K_t = \overline{\Sigma}_t C_t^T (C_t \overline{\Sigma}_t C_t^T + R_t)^{-1}$$

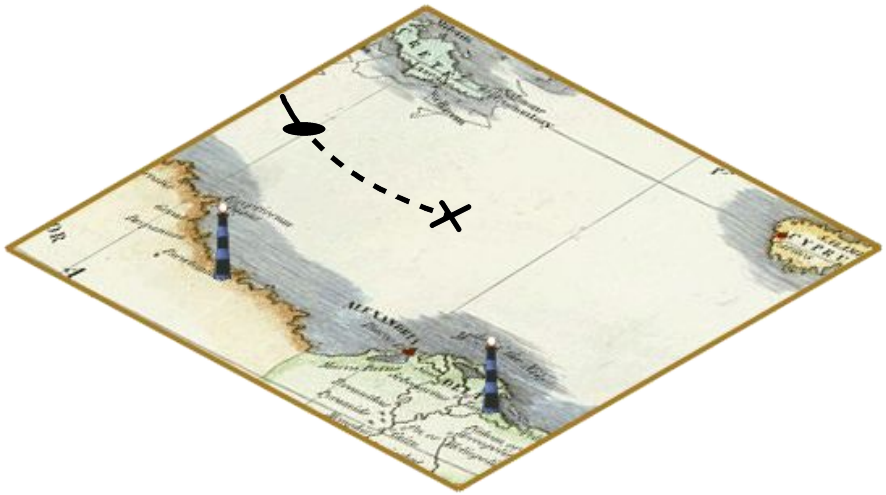
Algoritmo de filtro de Kalman

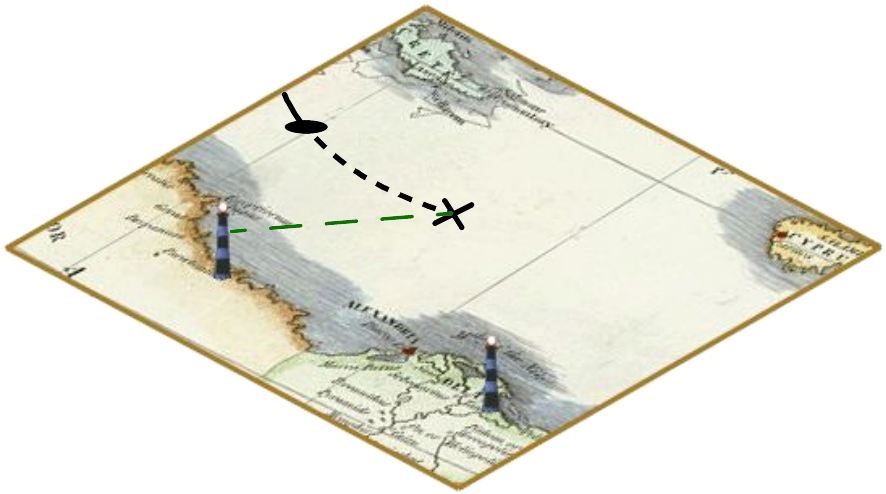
1. Algoritmo **Kalman_filter**(μ_{t-1}, Σ_{t-1}, u_t, z_t):

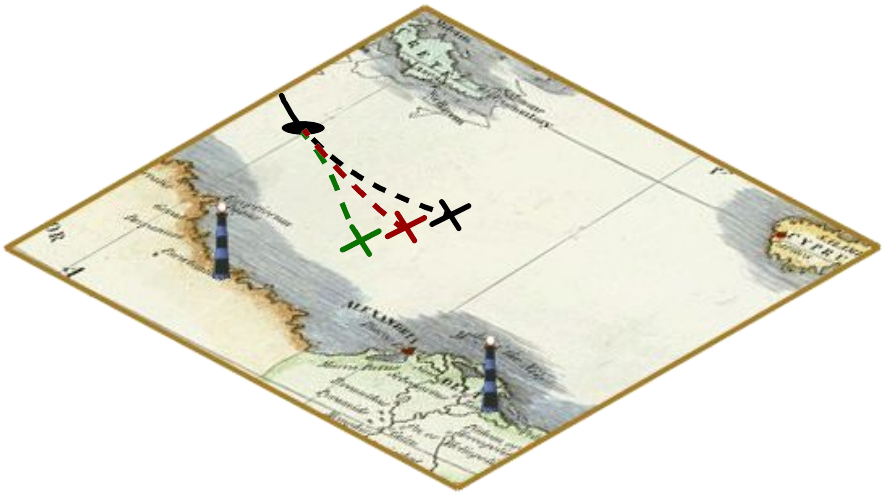
2. Predicción:
3. $\bar{\mu}_t = A_t \mu_{t-1} + B_t u_t$
4. $\bar{\Sigma}_t = A_t \Sigma_{t-1} A_t^T + Q_t$

5. Corrección:
6. $K_t = \bar{\Sigma}_t C_t^T (C_t \bar{\Sigma}_t C_t^T + R_t)^{-1}$
7. $\mu_t = \bar{\mu}_t + K_t (z_t - C_t \bar{\mu}_t)$
8. $\Sigma_t = (I - K_t C_t) \bar{\Sigma}_t$
9. Return μ_t, Σ_t

Algoritmo de filtro de Kalman

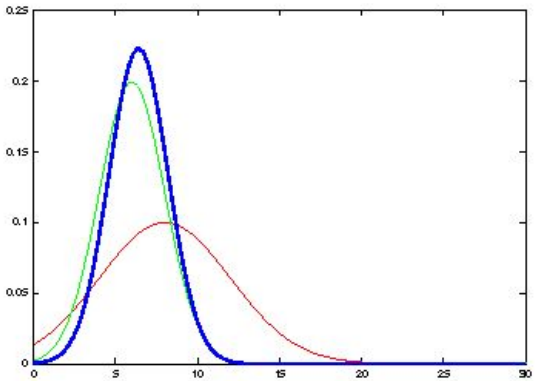






Ciclo predicción-Corrección

Predicción



$$\overline{bel}(x_t) = \begin{cases} \bar{\mu}_t = a_t \mu_{t-1} + b_t u_t \\ \bar{\sigma}_t^2 = a_t^2 \sigma_t^2 + \sigma_{act,t}^2 \end{cases}$$

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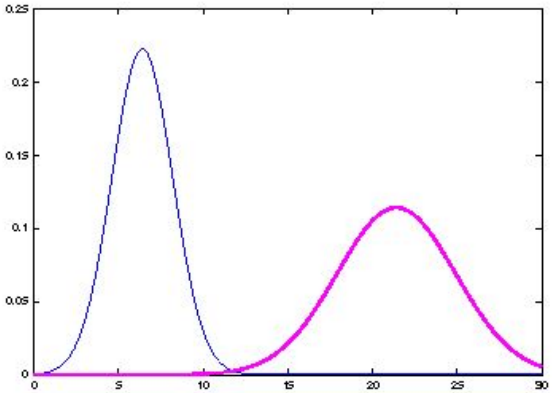
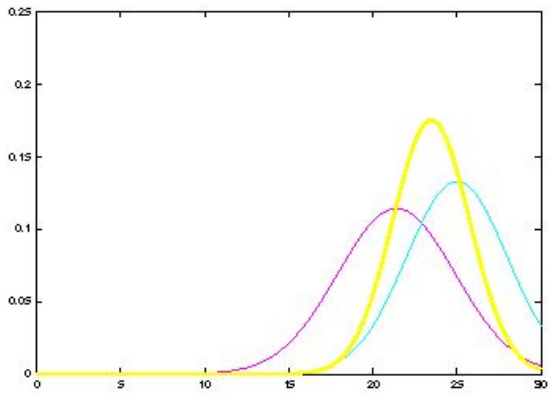


Ciclo predicción-Corrección



$$bel(x_t) = \begin{cases} \mu_t = \bar{\mu}_t + K_t(z_t - \bar{\mu}_t) \\ \sigma_t^2 = (1 - K_t)\bar{\sigma}_t^2 \end{cases}, K_t = \frac{\bar{\sigma}_t^2}{\bar{\sigma}_t^2 + \bar{\sigma}_{obs,t}^2}$$

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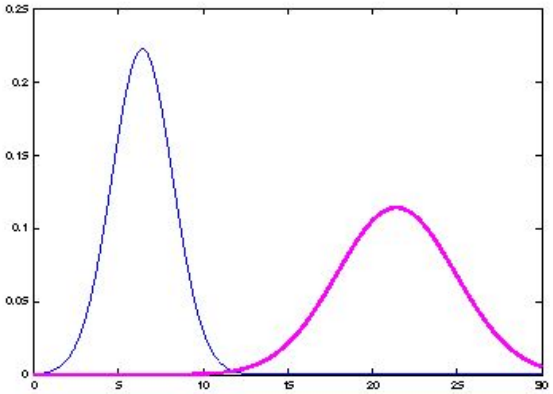

Corrección

Ciclo predicción-Corrección

Predicción

$$\overline{bel}(x_t) = \begin{cases} \overline{\mu}_t = a_t \mu_{t-1} + b_t u_t \\ \overline{\sigma}_t^2 = a_t^2 \sigma_t^2 + \sigma_{act,t}^2 \end{cases}$$

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$$bel(x_t) = \begin{cases} \mu_t = \overline{\mu}_t + K_t (z_t - \overline{\mu}_t) \\ \sigma_t^2 = (1 - K_t)\overline{\sigma}_t^2 \end{cases}, K_t = \frac{\overline{\sigma}_t^2}{\overline{\sigma}_t^2 + \overline{\sigma}_{obs,t}^2}$$

$$bel(x_t) = \begin{cases} \mu_t = \overline{\mu}_t + K_t (z_t - C_t \overline{\mu}_t) \\ \Sigma_t = (I - K_t C_t)\overline{\Sigma}_t \end{cases}, K_t = \overline{\Sigma}_t C_t^T (C_t \overline{\Sigma}_t C_t^T + R_t)^{-1}$$

Corrección

Resumen del filtro de Kalman

- Dos parámetros describen el belief del estado del sistema
- **Muy eficiente:** Polinómico en la dimensión de la medición k y la dimensión del estado n:

$$O(k^{2.376} + n^2)$$

- **Óptimo para sistemas Gaussianos lineales**!
- Pero: La mayoría de los sistemas robóticos son **no-lineales**!
- Sólo puede modelar beliefs unimodales